The value of energy efficiency in Tunisia's power system

What each MWh saved avoids in 2030, under the 35% renewable target

Rayen Romdhani, Energy Engineering, ENIM (Monastir, Tunisia)

PyPSA-Earth capacity-expansion model, 2 nodes, 8,760 hours

Each MWh saved avoids about 60 EUR of system cost

Saving 15% of 2030 demand in every hour (S1-15 against the base case S0), per MWh saved:

60 EUR

of system cost (investment + operation)

0.91 MWh

of gas burnt for power

0.18 t

of CO2

449 MW

of new capacity per TWh saved per year

The avoided cost is close to the average power price (55 EUR/MWh) plus the gas turbines the saving makes unnecessary.

Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

An open model, calibrated to 2024, then solved for 2030

Base network

PyPSA-Earth builds Tunisia's grid, plants, solar and wind potential and hourly demand from open data

2024 calibration

Official ONEM fleet; one efficiency factor fitted so gas burnt matches STEG within 0.4%

11 scenarios

Demand cut 10%-20%, cooling-first saving, shiftable load, gas price x0.5 and x1.5

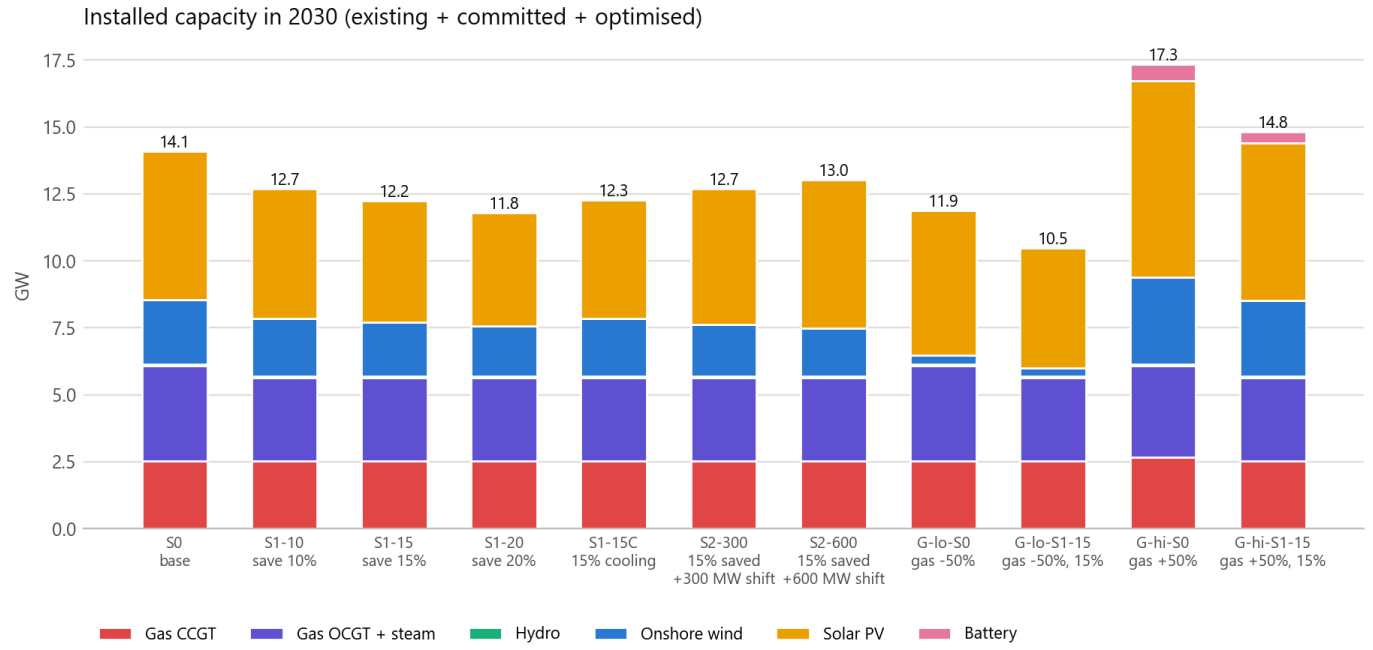
Least-cost 2030

New solar, wind, batteries and gas; at least 35% renewable and 10% reserve in every run

Every input number has a row and a source in the assumptions register. All 11 runs pass the post-solve checks.

Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

A 10% saving removes the need for new gas plant



Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

423 MW

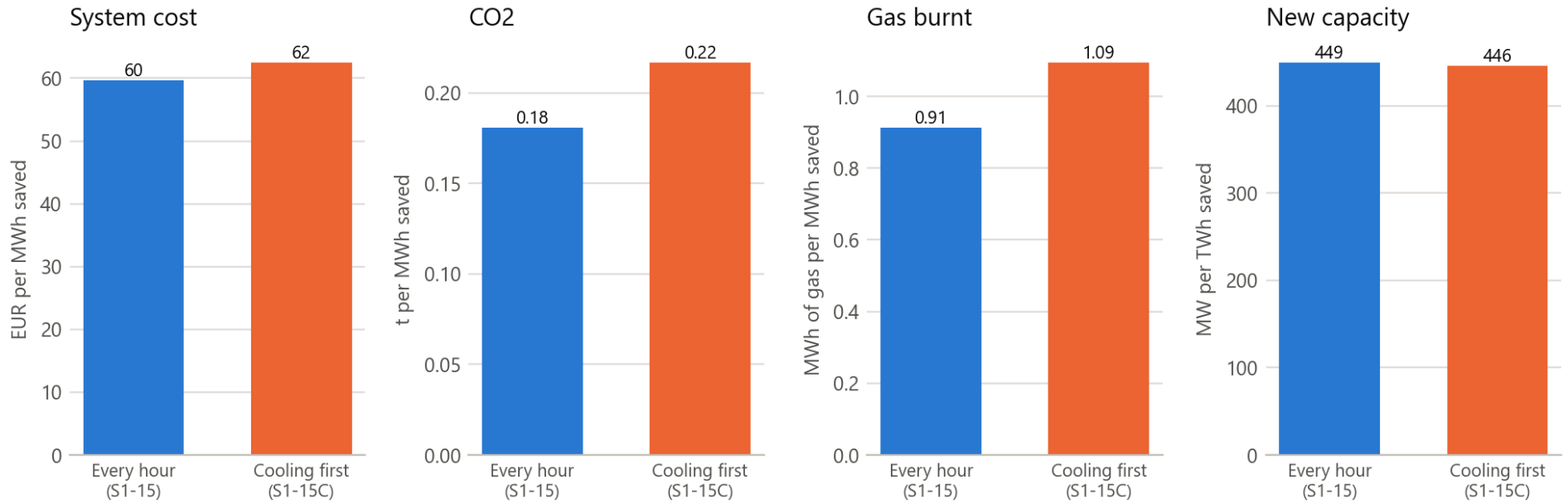
of new gas turbines in the base case, to keep a 10% reserve above the peak

Every efficiency scenario avoids all of it. Further savings then avoid mostly solar and wind that would otherwise be built.

Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

Saving during air-conditioning hours is worth more

Saving 15% evenly vs from summer cooling first (per MWh saved, vs S0)

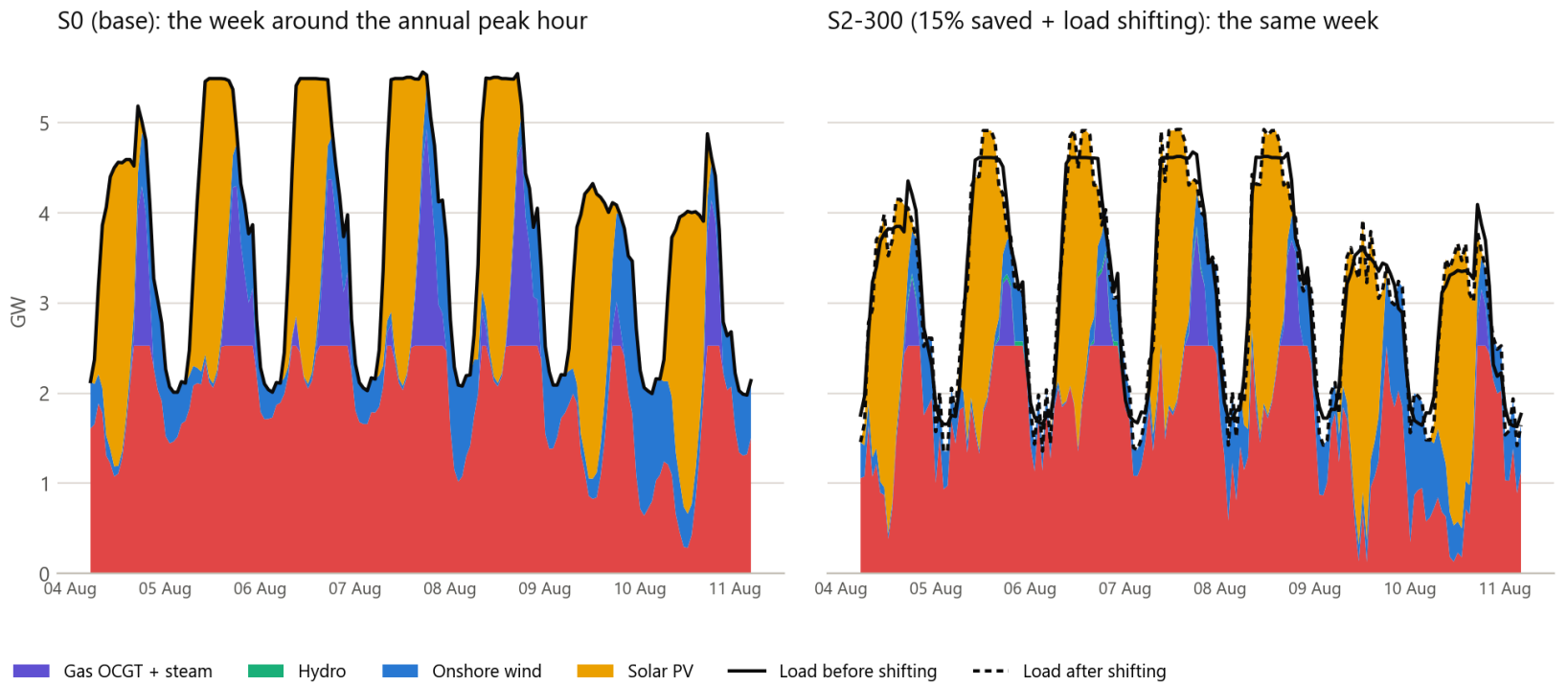


Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

Only 39% of the 15% saving fits on the summer cooling load, yet it avoids 20% more gas and CO2 per MWh than an even cut: those are the hours when gas runs.

Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

Shifting load by up to 3 hours adds to efficiency



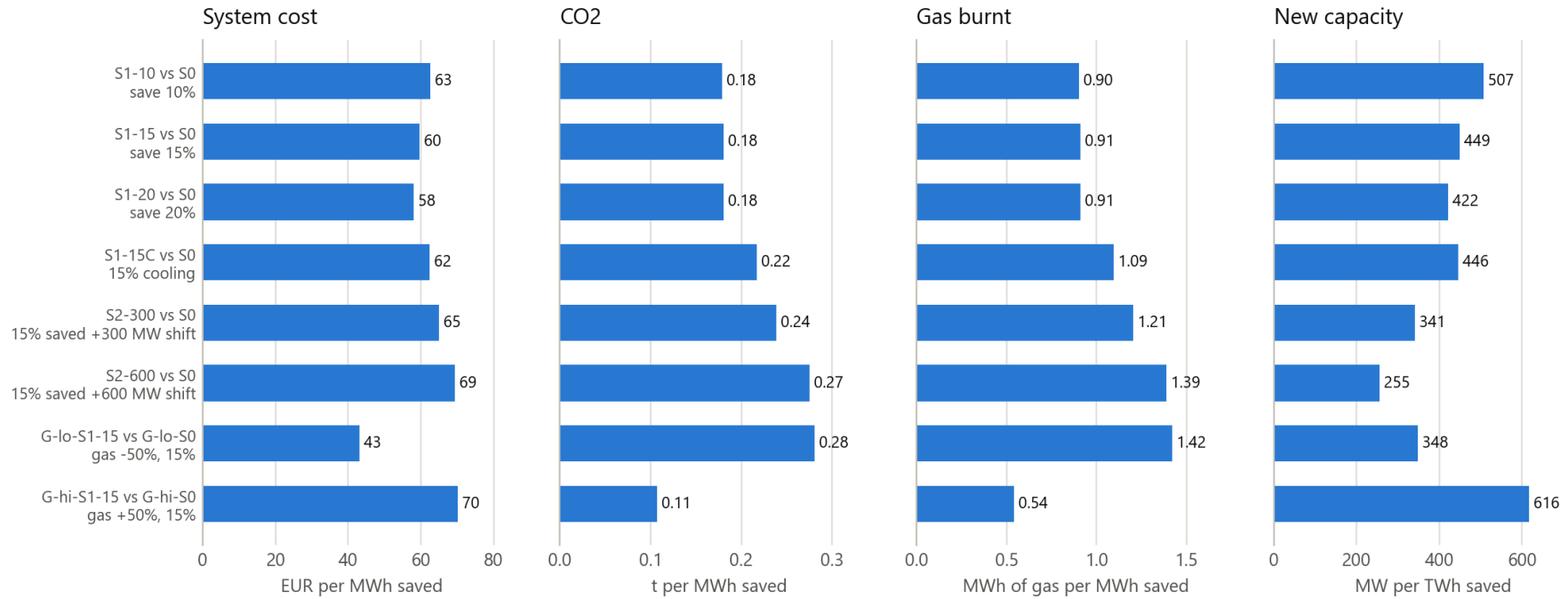
Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx; weather year 2013, times in UTC

300 MW of shiftable load saves another 22 M EUR and 0.24 Mt of CO2 per year, by moving morning and evening load into midday solar hours.

Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

The gas price decides what a saved MWh displaces

What each MWh of electricity saved avoids in 2030

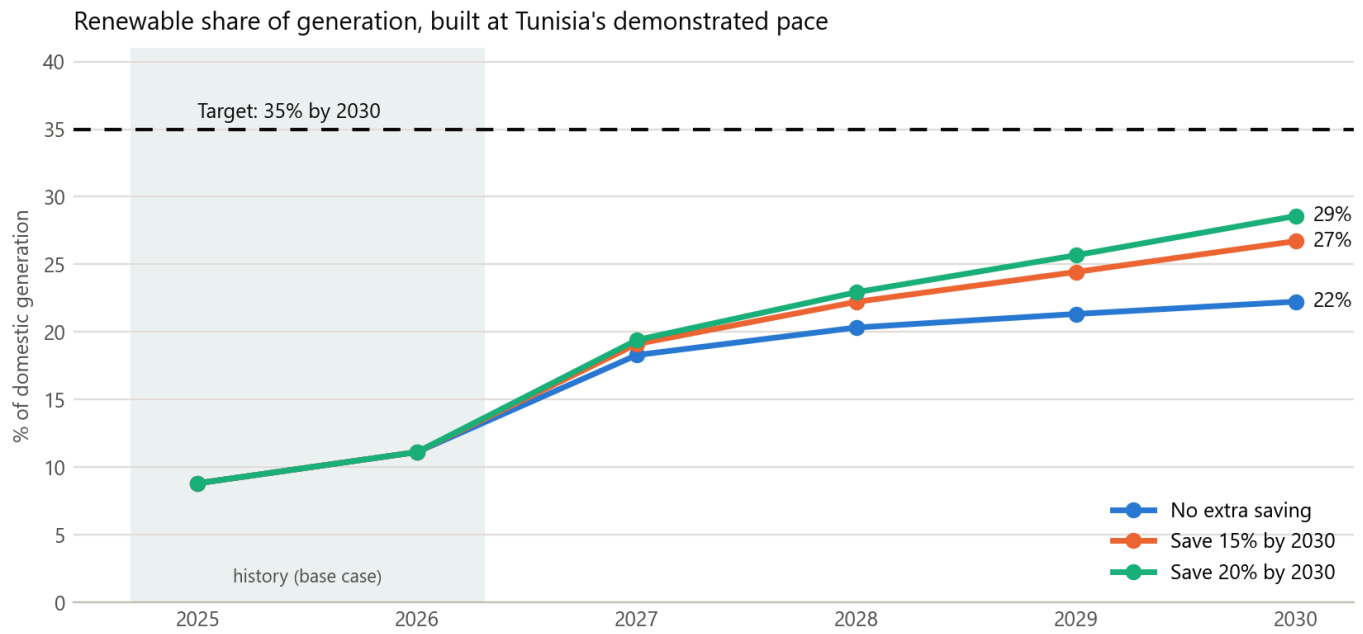


Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx; each scenario is compared with its reference at the same gas price

Cheap gas: each MWh saved avoids 0.28 t of CO2. Expensive gas: the system is already 70% renewable, so it avoids only 0.11 t of CO2 but 70 EUR of cost.

Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

At Tunisia's real pace, 35% is out of reach



Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx; build limits in the register (pathway_*)

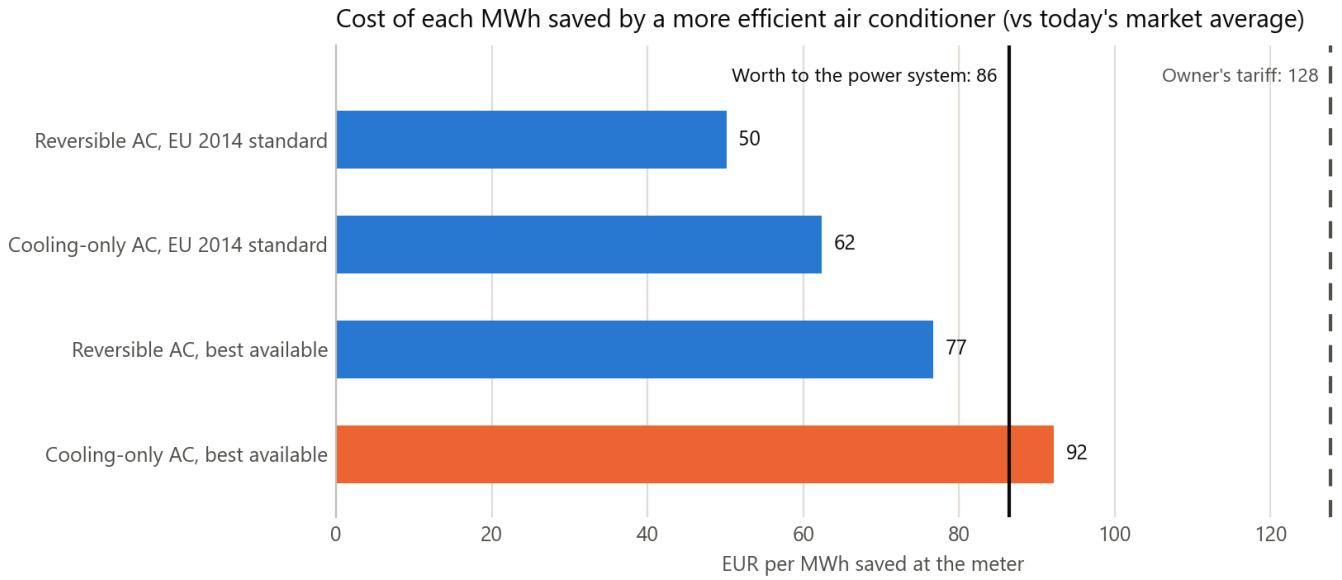
22%

renewables in 2030 without extra saving; 27% when saving 15%

Each MWh saved over 2027-2030 avoids 0.39 t of CO2 and 70 EUR: it comes off a gas plant.

Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

A stricter air-conditioner standard saves for less than it avoids



Source: LBNL 2023 (prices and kWh per unit); value: own model, S1-15C 2027-2030 plus 18% grid losses; results/saving_cost.csv

50-62 EUR

per MWh saved at the EU 2014 level, against 86 EUR avoided

It covers 9%-25% of the 15% saving; the other measures are not priced yet.

Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

What the numbers mean, and what comes next

Read with care

Not a forecast: 2030 alone has no build limit; the pathway's limits are judgements

Only air-conditioner standards are priced: the cost of the whole saving is unknown

One weather year; imports flat at their 2024 level

Demand growth from STEG's forecast as quoted by JICA (2022)

Next steps

Add the ELMED link to Italy as a scenario

Test faster build rates as the 2 GW wind tenders progress

Price lighting, building and industry savings in Tunisia

Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx